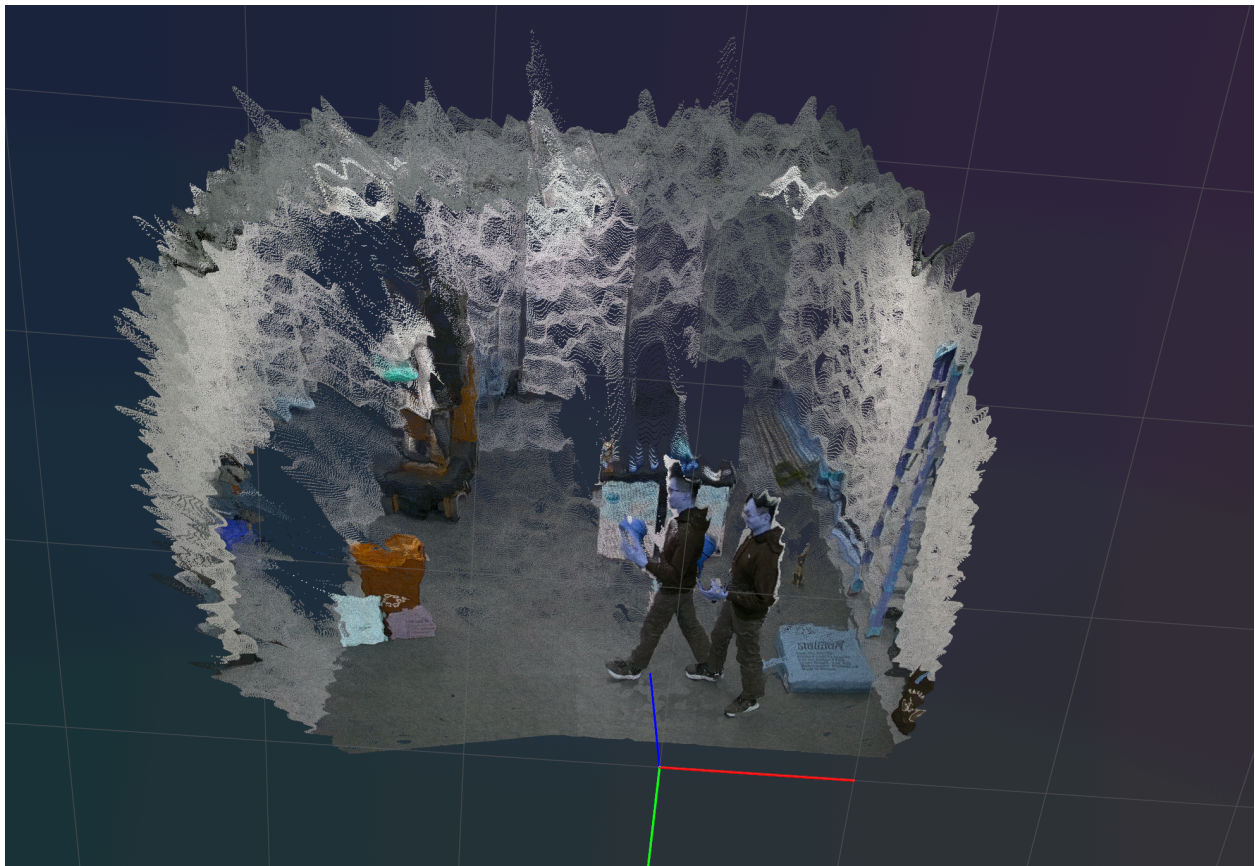


Steps in order.

Time Stamp 30, 60 - with poses estimated from pyslam - online

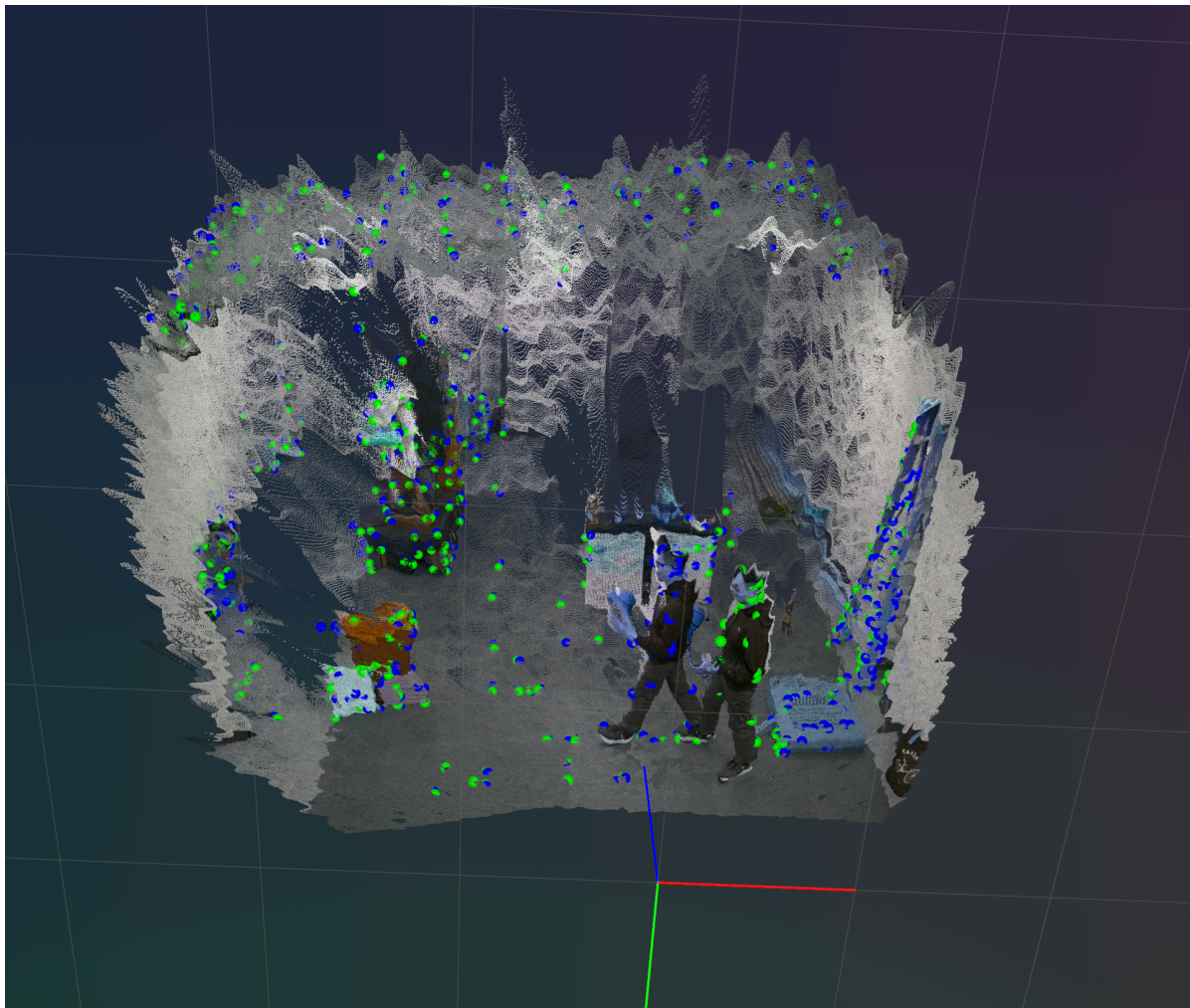


Two point clouds registered with their estimated poses

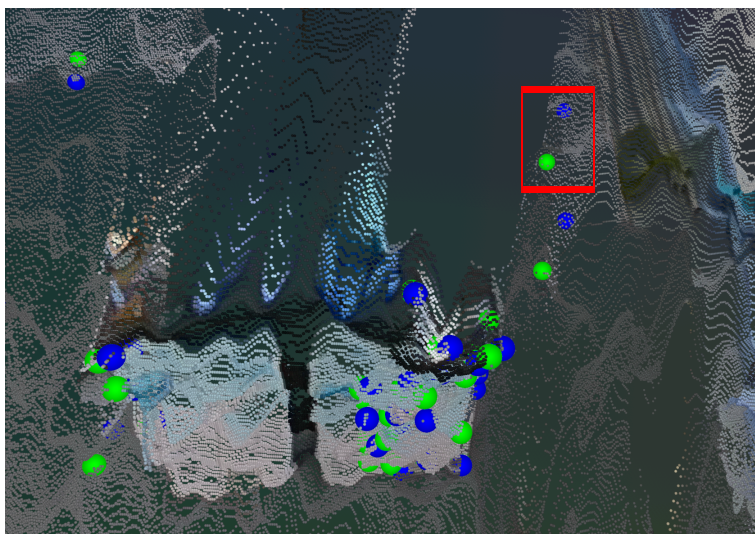


Remark: Depth is noisy for this dataset

Visualizing Matched Keypoints in 3D



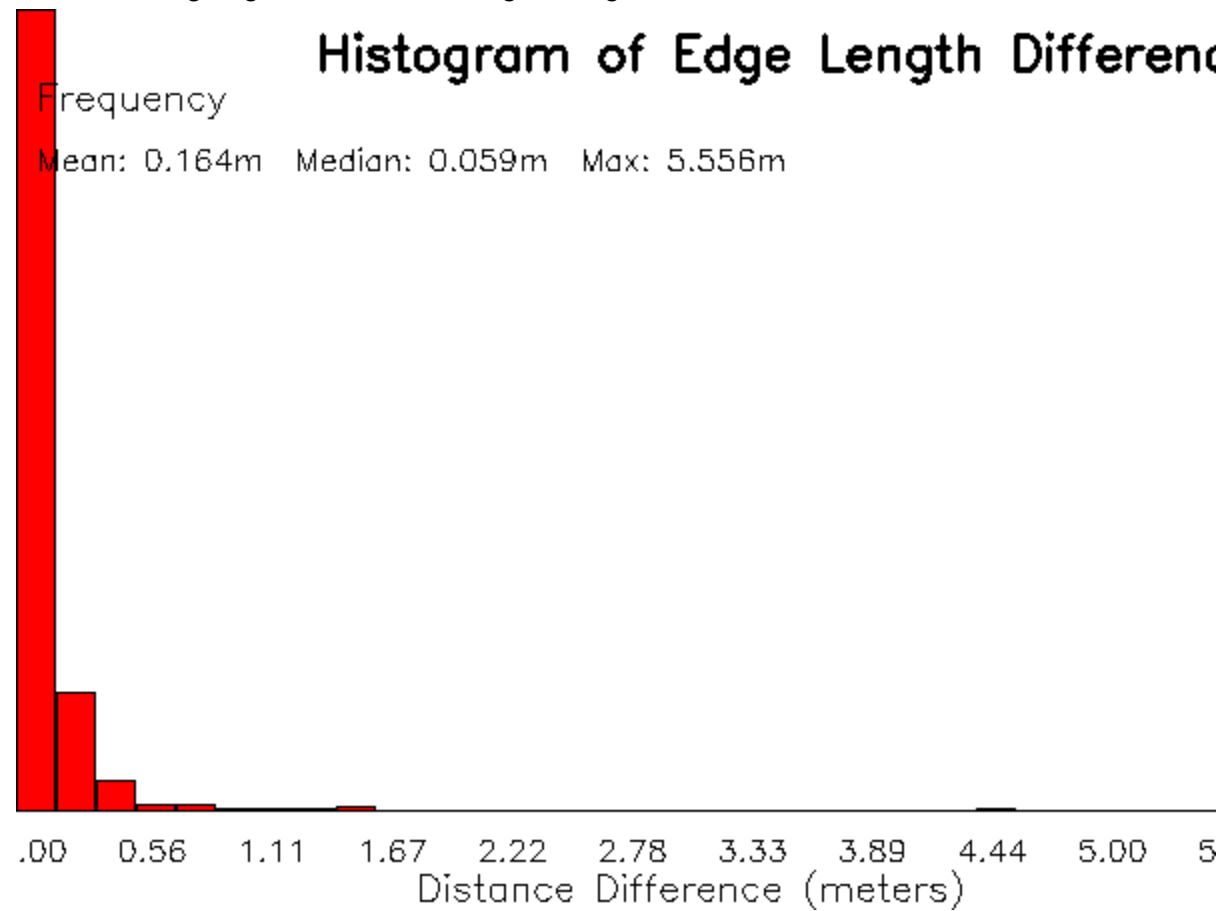
Remarks: Because of the noisy depth - some matched points have large displacements like shown below



A photograph of a man in a dark hoodie and glasses holding a basketball, standing within a large, green, wireframe mesh structure that resembles a virtual environment or a sensor field. The setting is an indoor room with a white wall, a wooden ladder, and various objects like a blue recycling bin and a table with fruit. The mesh is composed of numerous interconnected green lines forming a complex, irregular shape that encloses the man and the basketball.

[illegible]

Understanding edge differences through histogram



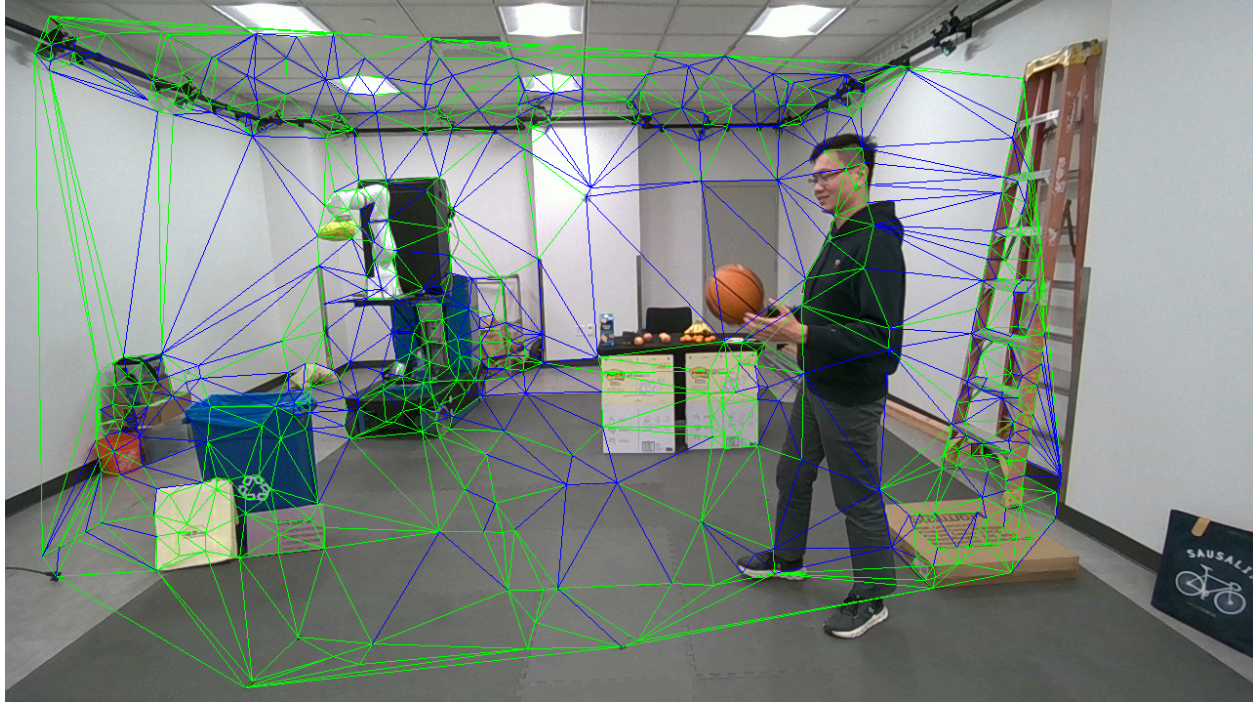
Results:

Mean difference ~ 16cm

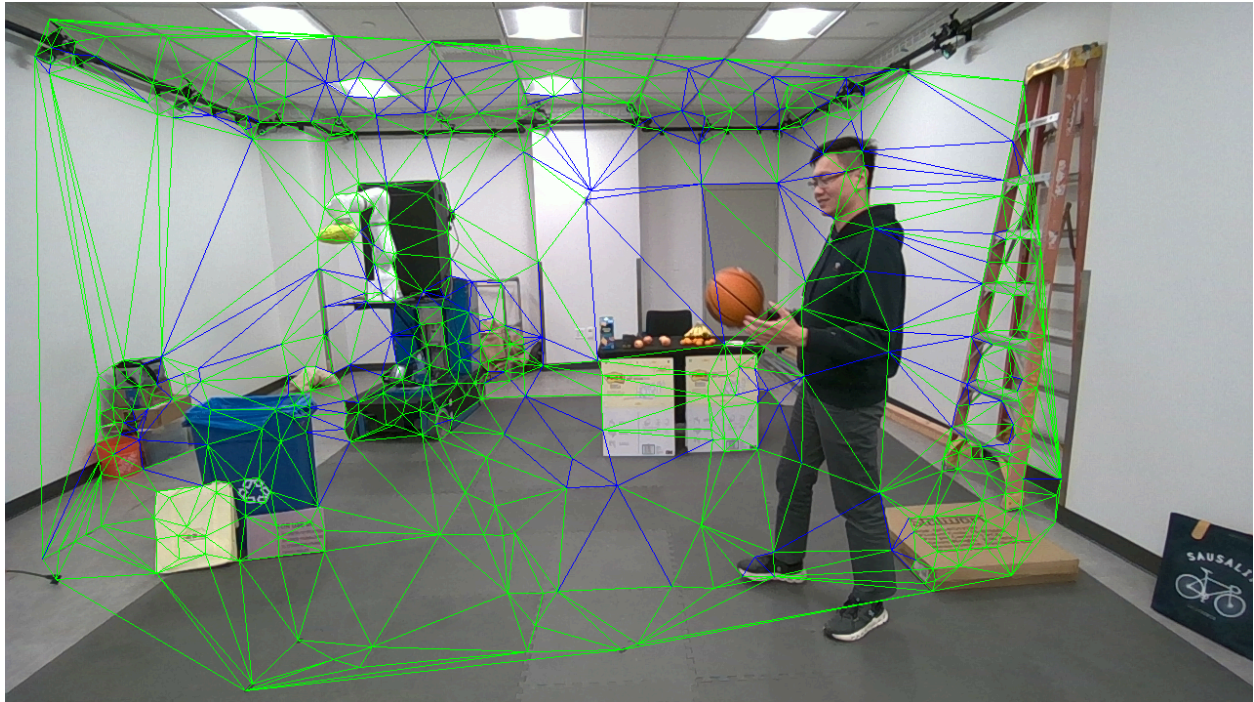
Median difference ~ 6cm

Results for difference edge thresholds:
BLUE EDGE is a dynamic edge
GREEN EDGE is a static edge

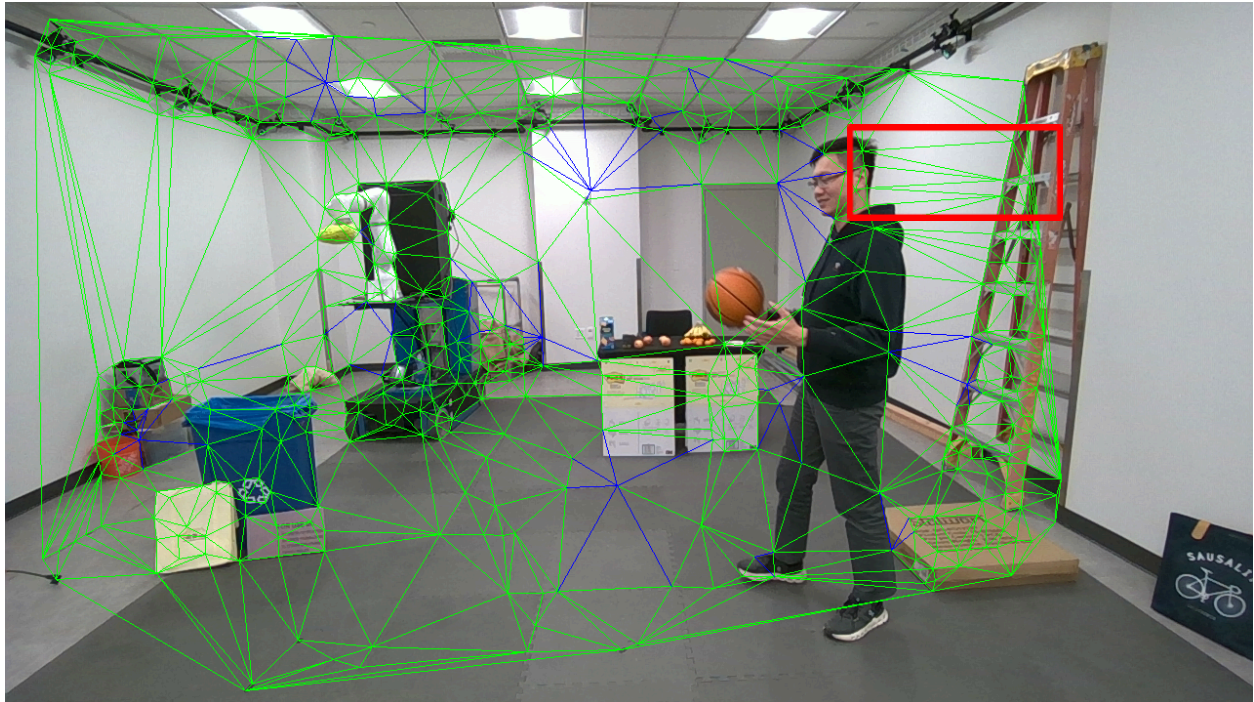
Threshold=10cm



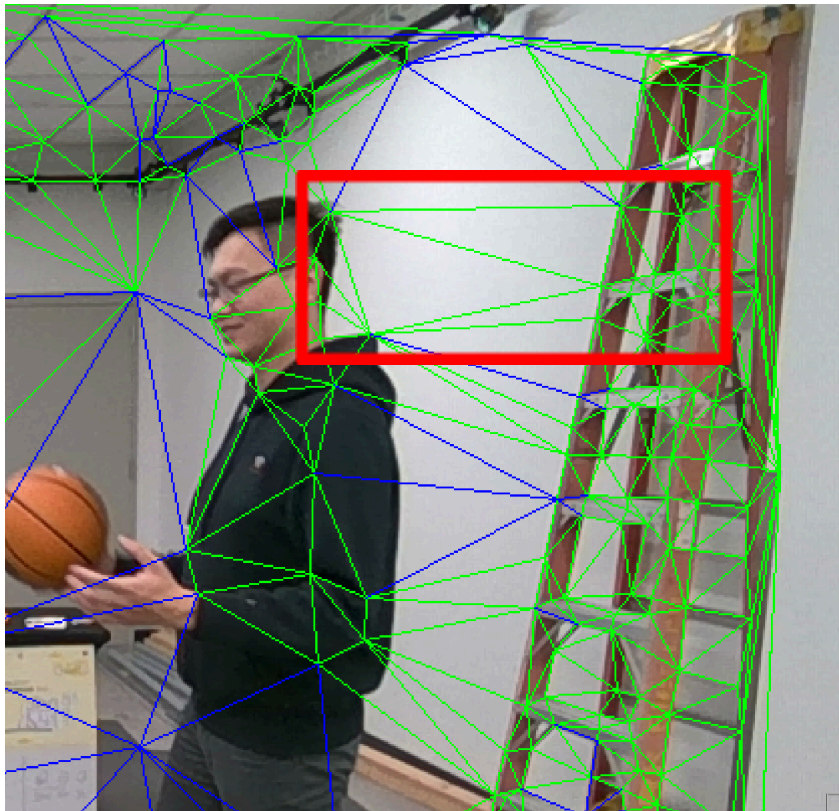
Threshold=20cm



Threshold = 40cm

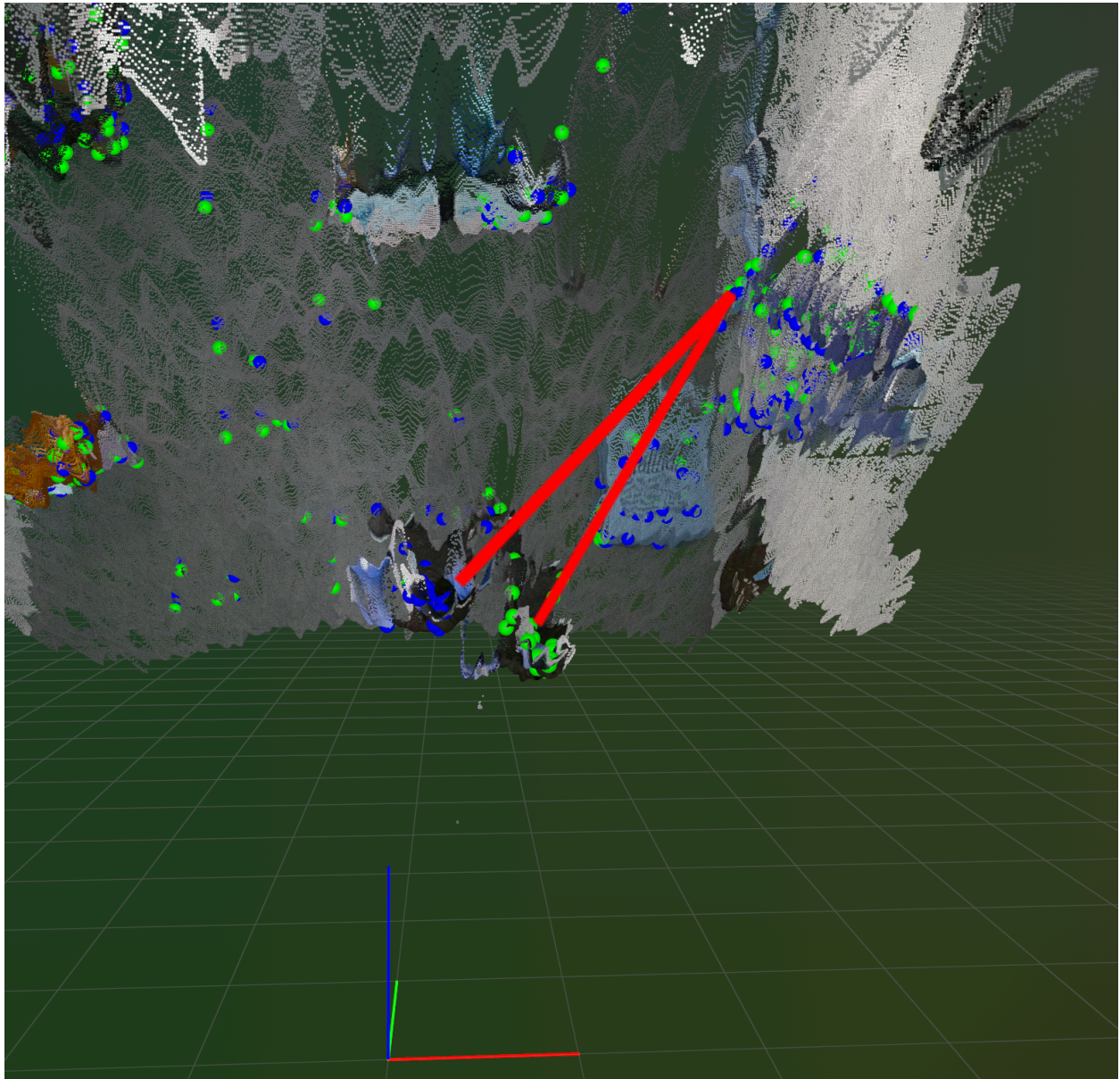


Remarks: At higher thresholds - effects on noise on static edges is negligible and hence they are detected static with more accuracy. But some actually dynamic edges (i.e. as shown in the red box) are still considered dynamic because those edge lengths are not changing.



Extended Remarks:

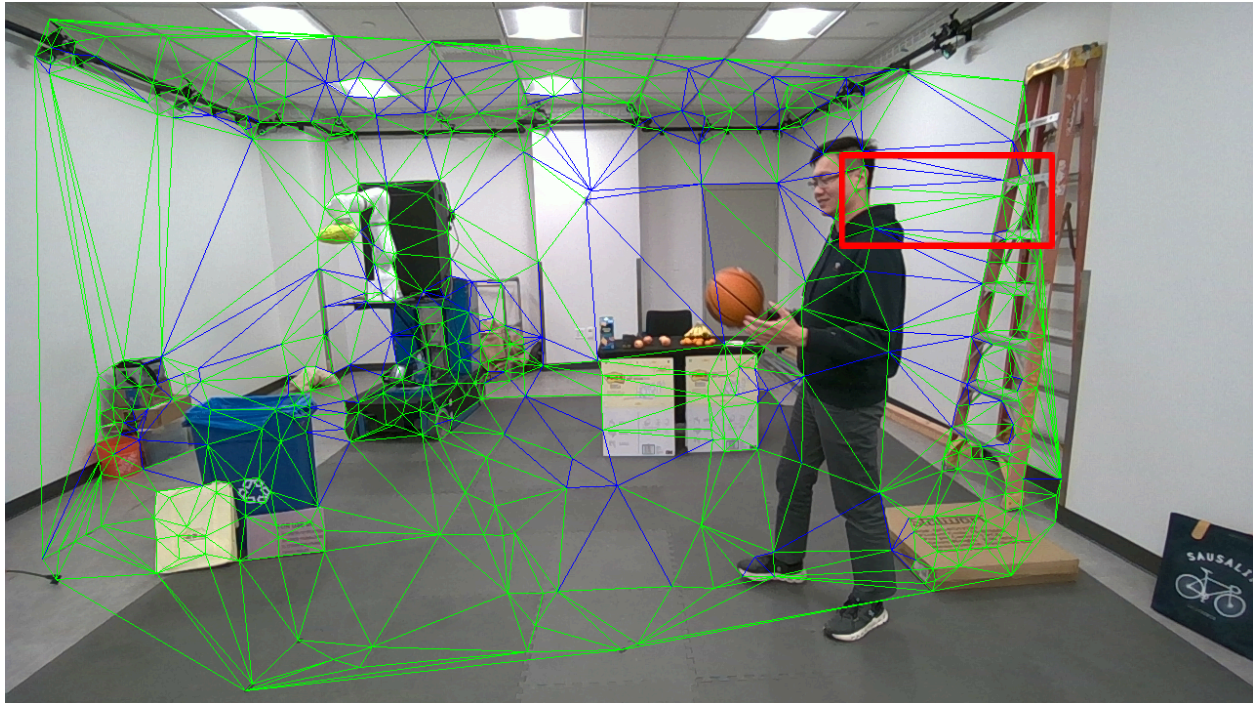
I think this is the case because the edges are rotated in 3D with the anchor as the static keypoint and the other keypoint on the circumference.



Remarks: I think, I should consider the case when the length doesn't change much, but the angle of the edge changes in 3D space then calculating that angle change also tells about the motion of the edge. Especially, when the edge length is large as shown in the image.

FINAL REMARKS:

Results of threshold = 20cm



Results of threshold = 20cm - this result looks promising given I can fix noisy depth by smoothing somehow - so I can remove all the false BLUEs(static edges but shown in Blue here) and at the same time, if I can somehow account for the change in the angle of the long edges (i.e. edges in the red box here -> false GREENs -> Falsely estimated as static because of rotation only motion in 3D) we can remove false negatives.

Finally, Improvement strategy for

- **False Negative** - Accounting rotation in the edges
- **False Positive** - Accounting for noise and also see if I can bring local map her, for example if the kp is available in local map, use local maps position, else use RGBD depth